Miraslau Pisaryk

Java / Spring Boot Backend Engineer | Distributed Microservices & AI Systems

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github.com/CHIKOJgg | linkedin.com/in/miroslav-pisaryk-953490261 | **100% Remote Worldwide | Open to B2B / Contractor Contracts**

PROFESSIONAL SUMMARY

Backend engineer specializing in distributed systems, event-driven microservices, and AI integrations on Java 21/25 and Spring Boot 3/4. Demonstrated track record designing high-throughput transactional backends (thread-safe in-memory order matching engine, real-time WebSocket depth streaming, Redis caching) and architecting autonomous AI agent platforms with Spring AI, Keycloak 26 (OAuth2/OIDC), RabbitMQ, and PostgreSQL 17 + pgvector RAG. Ensuring 100% integration test isolation via Testcontainers. English C1 Advanced; ready for immediate full-time remote B2B collaboration worldwide.

TECHNICAL SKILLS

Languages & Core: Java (17, 21, 25), SQL, Concurrency (Virtual Threads / Loom, synchronized, volatile, CompletableFuture, ConcurrentHashMap), Stream API, Generics, OOP

Microservices & Frameworks: Spring Boot 3.4, Spring AI, Spring Cloud OpenFeign, Spring Data JPA / Hibernate, Spring Security, WebSocket (STOMP), Resilience4j, Bucket4j, MapStruct, Lombok

IAM & Security: Keycloak 26 (OAuth2 Resource Server, OIDC, PKCE, JWT signature validation via JWKS, JIT User Provisioning, TokenRelayInterceptor)

Messaging & Broker: RabbitMQ 3.13 (Topic Exchanges, AMQP, JSON Serialization, Dead-Letter Queues, Decoupled Domain Events)

Databases & Vector Search: PostgreSQL 17, pgvector (Vector Embeddings, Cosine Similarity RAG), Redis 7 (Distributed Caching, Pub/Sub, Rate Limiting), Flyway Migrations, EXPLAIN ANALYZE

AI & LLM Integration: Spring AI (ChatClient, @Tool Function Calling, Dynamic RAG Prompts), Anthropic Claude API, Model Context Protocol (MCP)

DevOps & Testing: Docker, Docker Compose, GitHub Actions (CI/CD), Maven, Linux / Bash, JUnit 5, AssertJ, Mockito, Testcontainers (PostgreSQL, Redis, RabbitMQ)

Architecture & Standards: Distributed Microservices, Multi-Module Monorepo, Event-Driven Architecture, DDD, Clean Architecture, REST API RFC 7807, Agile (Scrum/Kanban)

Spoken Languages: English — C1 Advanced (EF SET 70/100, verified); Polish — beginner

EXPERIENCE & PRODUCTION-GRADE SYSTEMS

Nexgen Soft — Java Backend Developer

Java 21, Spring Boot 3.3, PostgreSQL, Redis, WebSocket (STOMP), Spring Security, Flyway, Docker, GitHub Actions

September 2025 – Present | Remote | Outsourcing & Distributed Development, FinTech Product Team

- **FinTech Platform Engineering:** Developing the backend core of a multi-currency banking and exchange platform for account operations, transaction processing, and live market quotes in an Agile team of 8.
- **In-Memory Matching Engine:** Architected and implemented a thread-safe in-memory order matching engine (price-time priority) for limit and market orders using Java Concurrency lock-free primitives and concurrent data structures.
- **Redis Caching & Load Reduction:** Integrated Redis for high-speed order book depth caching and message queuing, decreasing PostgreSQL transactional read load by ~40% during peak market volatility.
- **Real-Time WebSocket Streaming:** Engineered real-time order book and private account notification streaming over WebSocket (STOMP) for 200+ concurrent clients with sub-second delivery.
- **Financial Schema & Migrations:** Designed a normalized 13-table PostgreSQL schema with double-entry accounting ledgers and immutable audit trails; automated multi-stage schema evolution via Flyway.
- **Security & Testcontainers:** Enforced Spring Security with JWT token rotation; covered critical transactional code with JUnit 5 and Testcontainers (PostgreSQL, Redis), ensuring zero flaky integration tests.

AI Assistant Klawa — Software Engineer (Distributed Microservices & AI Engineering)

Java 25 (Virtual Threads), Spring Boot 3.4, Spring AI, PostgreSQL 17 + pgvector, Keycloak 26, RabbitMQ 3, Spring Cloud OpenFeign, Redis 7, Testcontainers

April 2025 – September 2025 | Remote | Multi-Module Maven Microservices Monorepo

- **Microservices Monorepo Migration:** Architected and executed the end-to-end migration from a monolithic prototype to an Event-Driven Microservices Architecture structured as a Maven multi-module monorepo (klawa-contracts, klawa-core, klawa-ai-agent).
- **Core Business Service & Virtual Threads:** Developed the high-throughput klawa-core service on Java 25 utilizing Virtual Threads (Project Loom) for non-blocking I/O execution across task orchestration, scheduling, and entity management.
- **Enterprise Keycloak 26 Security:** Integrated Keycloak 26 IAM as an OAuth2 Resource Server; implemented stateless JWT signature verification via JWKS, realm role mapping, and Just-In-Time (JIT) user provisioning.
- **Spring AI & pgvector RAG:** Built the dedicated klawa-ai-agent microservice on Spring AI; implemented Retrieval-Augmented Generation (RAG) using PostgreSQL 17 with pgvector extension for real-time semantic similarity search.
- **Autonomous Function Calling:** Implemented @Tool Function Calling allowing the LLM to autonomously trigger task creation and scheduling commands, executed synchronously in klawa-core via Spring Cloud OpenFeign with TokenRelayInterceptor (JWT propagation).
- **RabbitMQ Event-Driven Streaming:** Configured RabbitMQ 3.13 (Topic Exchange klawa.events) for asynchronous domain events (TaskCreatedEvent, AiTaskAnalysisRequestedEvent); decoupled vector embedding generation so HTTP transactions remain sub-50ms.
- **Resilience & Test Isolation:** Configured Resilience4j Circuit Breakers for AI vendor failover, Bucket4j token bucket rate limiting, Redis caching, and 100% isolated integration tests using Testcontainers (PostgreSQL, Redis, RabbitMQ).

ADDITIONAL QUANTITATIVE PROJECTS

Crypto Portfolio Optimizer — Quantitative Analysis & Backtesting Engine

Java 21, JavaFX, ojalgo, CoinGecko API, Maven | github.com/CHIKOJgg/MarkowitzModel-java-crypto-analysis

- **Algorithmic & Portfolio Models:** Engineered 12 quantitative alpha models (Momentum, EWMA, RSI, MACD, Bollinger) and 9 allocation frameworks (Markowitz Mean-Variance, Risk Parity, HRP, CVaR, Black-Litterman).
- **Backtesting & Stress Testing:** Built a walk-forward backtesting engine with Monte Carlo simulation (Geometric Brownian Motion) and Brinson attribution, calculating real-time Sharpe, Sortino, Calmar, and VaR/CVaR 95% metrics.

EDUCATION & CREDENTIALS

Belarusian State University of Informatics and Radioelectronics (BSUIR)

Bachelor's Degree in Information Systems and Technologies (Computer Science) | 2024 – 2028 | Minsk, Belarus

Verified English Certificate: EF SET — Score 70/100, C1 Advanced (Official verification: cert.efset.org/ru/cAsbkU)